

Pavithra Uthrah R. K.

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Profile

Software Development Engineer with strong foundations in Data Structures, Algorithms, Object-Oriented Design, and System Design — backed by hands-on experience building scalable, production-grade applications, SaaS platforms, and distributed systems. Proficient in Java, Python, C++, and modern web technologies (Django, React, Node.js) with practical experience in microservices architecture, cloud platforms (Azure), containerization (Docker), and relational database design. Experienced across the complete SDLC — from requirements gathering and design to development, testing, deployment, and monitoring — in fast-paced, agile environments. Strong analytical, problem-solving, and debugging skills, demonstrated through internships and leadership of a 250+ member technical community. Passionate about building robust, maintainable software that delivers real business impact.

Education

Bachelor of Technology in Computer Science and Engineering

Aug 2023 – May 2027

Vellore Institute of Technology, Chennai

Higher Secondary Education (Class XII)

Graduated May 2023

ISC — St. Michael's Academy, Chennai

Technical Skills

- **Programming Languages:** Java, Python, C++, C, SQL, JavaScript/TypeScript, Shell Scripting
- **Core CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Design (OOD), Operating Systems, DBMS, Computer Networks, System Design, Complexity Analysis
- **Backend & Full-Stack:** Django, FastAPI, Node.js, Express, React, REST APIs, WebSocket, Microservices Architecture, Spring Boot (Basics)
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, SQL Server, Database Design, Query Optimization
- **Cloud & DevOps:** Microsoft Azure, Docker, Git/GitHub, Azure DevOps, Jenkins, CI/CD, Linux/Unix
- **Software Engineering:** SDLC, Agile/Scrum, Design Patterns, Code Reviews, Unit Testing, Test Automation, Debugging, Technical Documentation
- **AI & Data:** Generative AI, LLMs, RAG, Prompt Engineering, PySpark, Databricks, Power BI

Work Experience

Software Development Engineer – AIC Crescent Innovation & Incubation Council, Chennai

Internship | On-site | May 2026 – Jun 2026

- Architected and built a multi-tenant ERP backend skeleton supporting 14 entities and 140+ startup organizations using Django REST Framework, PostgreSQL, and Role-Based Access Control — demonstrating strong OOD and SDLC execution.
- Designed and implemented a facility-booking module with core scheduling logic, conflict detection, and REST API integration.
- Developed a WhatsApp-based workflow automation prototype using Python, REST APIs, and PostgreSQL in a 4-week Agile sprint; delivered as a production-ready solution adopted into daily operations.
- Participated in code reviews and unit test creation, contributing to software quality, maintainability, and coding standards.
- Coordinated delivery of 3 parallel workstreams, managing timelines and stakeholder syncs.
- Wrote comprehensive technical documentation and architecture notes to enable smooth handoff.

Data Engineer – Kaar Technologies, Chennai

Internship | On-site | Jul 2026 – Present

- Built an enterprise-scale EV Fleet Lakehouse Platform processing 50GB+ of vehicle telemetry through Medallion Architecture using Databricks, PySpark, and Delta Lake — applying software engineering best practices for scalable, maintainable code.
- Designed an AI-Powered SAP ERP Intelligence Assistant using Databricks Genie, RAG, Vector Search, and LLMs — enabling natural-language access to enterprise data.
- Architected an AI-Powered Multi-Agent Data Engineering Assistant combining Data Quality, Root Cause Analysis, and Predictive Failure Detection agents using LangGraph and Databricks AI/Genie.
- Collaborated with 6 cross-functional stakeholders (SAP, BI, Engineering) to translate business requirements into scalable technical solutions.
- Applied CI/CD workflows and version control (Git) for pipeline deployment and code management.

Projects

Women360 - Healthcare & Wellness SaaS

Cloud-Native SaaS, REST APIs, RBAC, Cloud Database, Responsive Web

- Built a full-scale cloud-native SaaS platform for women's health and wellness tracking with role-based access for Women, Wellness Coaches, and Admins.
- Architected secure data privacy controls for sensitive health information and designed REST APIs with cloud-hosted database and public deployment.
- Demonstrated end-to-end product development from concept to production deployment.

E-Commerce Platform – Full-Stack Microservices

React, Node.js, Express, PostgreSQL, Docker, Redis, Stripe, Nginx

- Built a production-grade e-commerce platform with microservices architecture including product catalog, order processing, payment, and user authentication services.
- Implemented JWT-based authentication, Redis caching for session management, and Stripe integration for payment processing.
- Deployed using Docker containers with Nginx as reverse proxy, achieving 99.9% uptime during load testing with 1000 concurrent users.
- Applied Object-Oriented Design principles, REST API best practices, and cloud-native deployment strategies.

AIC-CIIC Startup Incubation ERP

Django, PostgreSQL, Bootstrap, RBAC, REST APIs

- Architected and built a full-scale multi-tenant ERP managing complete startup lifecycle for 140+ organizations — applications, onboarding, lab/hall bookings, mentorship, services, finance/invoicing, and document management.
- Implemented Role-Based Access Control across 5 user personas and built calendar-based booking systems with conflict detection and automated notifications.
- Delivered comprehensive technical documentation and architecture notes for smooth handoff.

Real-Time Collaborative Document Editor

React, WebSocket, Node.js, MongoDB, Redis

- Engineered a Google Docs-style collaborative editor supporting real-time multi-user editing using WebSockets and operational transformation.
- Implemented document versioning, change history, and cursor synchronization with 50ms average latency.
- Designed the backend to support 200+ concurrent users with automatic conflict resolution.

Last-Mile Delivery Tracker

Backend API, Frontend, PostgreSQL, Rate Engine, Auto-Assignment

- Built a delivery management platform with auto-calculated charges based on zone detection, volumetric weight, B2B/B2C rate cards, and COD surcharge.
- Implemented intelligent auto-assignment logic to assign nearest available delivery agent and designed immutable order status lifecycle with full tracking history.
- Integrated email notifications for customers on every status change and built admin dashboard for order management.

Worklog Automation Platform (AI-Powered)

FastAPI, WhatsApp API, OpenAI/Gemini, PostgreSQL, Docker

- Built an AI-powered worklog automation platform that extracts, tracks, and reports employee daily updates from WhatsApp using OpenAI and Google Gemini APIs.
- Designed rule-based parser as fallback, reducing manual reporting effort by 75% and eliminating spreadsheet-based worklog management.
- Containerized the full stack with Docker and delivered as a production-ready solution adopted into daily operations.

Certifications

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| • Google Data Analytics Professional Certificate (v.3) | Issued by Google, January 2026. |
| • Data Science & Analytics Certificate | Issued by HP LIFE, August 2026. |

Leadership & Activities

- **Chairperson, Open Source Programming Club, VIT Chennai** (Apr 2025 – Present)
 - Lead a 250+ member technical community, driving strategy, operations, partnerships, and cross-functional initiatives. Organized 3 national-level hackathons and 10+ technical events, workshops, coding competitions, and speaker sessions.
 - Scaled the club to become the university's No. 1 technical club within two years of its launch, growing membership to 300+ and organizing 30+ national technical events.
- **Student Mentor Volunteer, U&I Trust, India** (Jul 2026 – Present)

- Provide weekly academic tutoring and holistic mentorship to primary school students, supporting their educational progress, confidence, life skills, and personal development through consistent guidance and positive role modelling.
- **Organizer – Spectrum, VOID.v1, Glytch, TechnoVIT’25, VIT Chennai**
 - Coordinated operations, participant engagement, and logistics for university technical festivals and hackathons involving 3000+ participants; served on the Organizing Committee for 10+ other technical events.
- **Rotaractor, Rotaract Club of VIT Chennai** (Jul 2025 – Jun 2026)
- **Management Member, Microsoft Innovations Club, VIT Chennai** (Sep 2024 – Apr 2025)